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Research paper

A dual-stream CNN-based ConvMixer for durian ripeness classification using magnitude and phase features from knocking sounds

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ABSTRACT

Knocking-sound analysis provides a non-destructive method for assessing durian ripeness; however, most convolutional neural network methods mainly emphasize magnitude information while disregarding phase information that reflects the overall signal structure. KnockNet is proposed as a model that processes the magnitude and phase feature domains and integrates them to exploit the complementary benefits of both representations. Mel-frequency cepstral coefficients contain the power distribution across frequency bands, while modified group delay cepstral coefficients capture delicate phase dynamics. Each feature stream is processed through specialized branches using a local channel attention mechanism, which emphasizes significant spectral channels, and a streamlined convolutional mixer block captures global context (broader temporal–spectral patterns), while convolutional layers with local channel attention capture local context (fine-grained spectral details). The resulting embeddings are combined to improve interpretation of the knocking event. We collected a dataset of 52 Monthong durian fruits in Chanthaburi Province, Thailand, and evaluated the model using five-fold cross-validation. KnockNet reached 96.34 ± 0.40% accuracy (Precision/Recall/F1-score: 96.44/96.34/96.34%), exceeding mel-frequency cepstral coefficient-only (95.48 ± 0.61%), CNN (94.21 ± 0.81%), convolutional neural network-long short-term temory (94.82 ± 1.13%), and support vector machine (93.83 ± 0.83%). These findings confirm that the combined use of magnitude and phase information, improved by focused attention and proper mixing, produces a practical and effective instrument for assessing durian maturity in the field.

Nomenclature

List of abbreviations

ANN	Artificial Neural Network.
AUC	Area Under the Receiver-Operating Characteristic (ROC) curve — discrimination metric for binary/multi-class cases.
BN	Batch Normalization.
CNN	Convolutional Neural Network.
CNN–LSTM	CNN-Long Short-Term Memory.
ConvMixer	Convolutional Mixer.
DCT	Discrete Cosine Transform.
DL	Deep Learning.
DM	Dry Matter (content) of the fruit pulp.
DNN	Deep Neural Network.
DWConv	Depthwise Convolution.
DT	Decision Trees.

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EER	Equal Error Rate.
EIS	Electrical Impedance Spectroscopy.
F1	F1-score — harmonic mean of precision and recall.
FC	Fully Connected — dense linear layer.
FFT	Fast Fourier Transform.
FLOPs	Floating-point operations.
GELU	Gaussian Error Linear Unit.
GD	Group Delay.
HW	Hardware.
KNN	k-Nearest Neighbors.
LCA	Local Channel Attention.
LDA	Linear Discriminant Analysis.
LSTM	Long Short-Term Memory.
MCU	Microcontroller Unit.
MFCC	Mel-Frequency Cepstral Coefficient.
MGDCC	Modified Group Delay Cepstral Coefficient.
ML	Machine Learning.
NB	Naïve Bayes.
NIR	Near-Infrared (spectroscopy).
PCA	Principal Component Analysis.
PCR	Principal Component Regression.
PLS	Partial Least Squares (regression).
PLS-DA	Partial Least Squares Discriminant Analysis.
PLSR	Partial Least Squares Regression.
RMSEP	Root Mean Square Error of Prediction.
PNCC	Power-Normalized Cepstral Coefficient.
PWConv	Pointwise Convolution.
RBF/rbf	Radial Basis Function
ReLU	Rectified Linear Unit.
SS	Single-Stream.
SS-CNN	Single-Stream CNN.
STFT	Short-Time Fourier Transform.
SVM	Support Vector Machine.
SWNIR	Short-Wave Near-Infrared.
Vis/NIR	Visible and Near-Infrared spectroscopy or imaging.

List of definitions

1×1 **projection**	Pointwise convolution used for channel mixing and dimensionality adjustment.
Channel-wise concat.	Concatenation performed along the channel dimension at the fusion stage.
Cross-validation	Stratified five-fold splitting of repetitions for training/validation/testing in experiments.
Curing	Post-harvest treatment applied to fruit to obtain the "Ripe" class.
DAA	Days After Anthesis; the number of days elapsed since durian flowering (bloom).
Dual-stream CNN	Two parallel CNN branches (MFCC and MGDCC) processed separately and fused at a late stage.
Dual-stream fusion	Late fusion via channel-wise concatenation, 1×1 projection, and patch embedding before ConvMixer.
DW%	Dry-weight percentage, defined as $100 \times W_{\mathrm{dry}}/W_{\mathrm{fresh}}$; Monthong maturity threshold $\geq 32\%$.
Efficiency	Model efficiency metrics: parameter count, FLOPs, stored size (float32 / int8), and single-instance (batch-1) latency.
Ensemble	Ensemble learning methods (e.g., random forests, boosting).
Embedding	Learned feature vector after fusion/ConvMixer used for downstream classification.
Global context	Coarser temporal–spectral dependencies captured after dual-stream fusion and ConvMixer processing.
KnockNet	The proposed dual-stream CNN with Local Channel Attention (LCA) and ConvMixer blocks, using MFCC and MGDCC features.
Local context	Fine-grained spectral patterns captured within each CNN branch (including LCA effects).
Maturity	Physiological stage determined by Days After Anthesis (DAA) and dry-weight percentage (DW%). Categories used: *Immature* (82–106 DAA), *Mature* (113–148 DAA and DW% $\geq 32\%$), and *Ripe* (post-harvest, ready-to-eat following curing).
Patch embedding	Strided convolution that converts feature maps into lower-resolution patch tokens for ConvMixer processing.
Performance	Reported metrics: Accuracy, Precision, Recall, F1-score (macro); values reported as mean ± SD across folds.
Repetition	A single ∼0.5 s knock trial (audio segment).
Ripeness	Ready-to-eat status after curing; classes used in this study: *Immature / Mature / Ripe*.

1. Introduction

Durian (*Durio zibethinus*), often known as the "king of fruits," is a significant agricultural product. Thailand is recognized as the largest producer and exporter, with durian exports substantially contributing to the country's agricultural economy [1]. In 2024, Thailand's total fruit exports reached $6.51 billion, with durian playing a pivotal role in driving this remarkable achievement [2].

The "Monthong" cultivar of durian is especially renowned due to its delightful flavor, thick flesh, mild aroma, and excellent shelf life [3]. Its high commercial value and consistent demand, particularly in international markets such as China, make it a key driver of agricultural export revenue for Thailand [4]. Beyond Thailand, other major countries producing durian including Malaysia, Indonesia, and Vietnam also make substantial contributions to the global durian supply, ensuring year-round availability in key markets.[1] Furthermore, the fruit's appeal is rapidly extending beyond Asia, with increasing demand observed in emerging markets such as Europe and the Middle East, where consumer interest in exotic tropical fruits is on the rise.[2,3] These distinctive qualities make Monthong durians highly attractive to consumers and processors, facilitating their transformation into various value-added products such as frozen durian, durian chips, and processed desserts, further enhancing their market potential and economic impact [3].

Evaluating durian ripeness accurately is essential to ensure that the fruit reaches consumers at optimal quality, thereby maximizing customer satisfaction and economic returns. Accurate ripeness assessment also reduces post-harvest losses and aids supply chain management by

[1] FFTC/FAO overview identifies Thailand, Malaysia, Vietnam, and Indonesia as the major durian-producing countries globally. See: Tai Wei Lim, "The Friendly Durian Competition in Southeast Asia," FFTC Agricultural Policy Platform, 18 Sep 2024. Available at: https://ap.fftc.org.tw/article/3641.

[2] European demand for exotic fruit is growing: imports from developing countries increased by 33% between 2018 and 2022. See: CBI (Netherlands), "The European market potential for exotic fruit," updated 23 Jan 2024. Available at: https://www.cbi.eu/market-information/fresh-fruit-vegetables/exotic-tropical-fruit/market-potential.

[3] Middle East demand indicators: in 2023 the United Arab Emirates imported $1.02 million of fresh durians (18th largest importer) and Saudi Arabia imported $438,000. Sources: OEC, "Fresh Durians in United Arab Emirates," https://oec.world/en/profile/bilateral-product/fresh-durians/reporter/are; and OEC, "Fresh Durians in Saudi Arabia," https://oec.world/en/profile/bilateral-product/fresh-durians/reporter/sau.

aligning harvest periods with market demands and export quality standards. Post-harvest studies report that tropical fruits including durian experience average losses of 15%–25% of total production,[4,5] while fruit rots alone can account for an additional 10%–25% loss.[6] However, traditional ripeness evaluation methods, which commonly rely on subjective sensory judgments or invasive techniques, often produce inconsistent and inefficient results. In response, non-invasive methods have emerged as promising alternatives, providing more consistent and objective classifications of ripeness [5].

Several non-invasive methods have been investigated to evaluate durian ripeness, which can be broadly categorized into vision-based, spectroscopy-based, sensor-based, and acoustic/vibration-based techniques [6–11]. Vision-based techniques [7,8] utilize Deep Learning (DL) frameworks, including Convolutional Neural Network (CNN) [7] and image [8], attaining near-optimal accuracy in controlled conditions; however, they often require significant computational power and uniform lighting conditions. Spectroscopy methods, such as Near-Infrared (NIR) [12] and thermal imaging [13], provide high precision in assessing internal fruit properties; however, they need costly equipment and rigorous calibration. Sensor-based methodologies, including electronic noses [14,15] and Electrical Impedance Spectroscopy (EIS) [16], concentrate on identifying volatile substances or alterations in tissue impedance. These non-invasive approaches are systematically summarized in Table 1. While these can accurately differentiate ripeness stages, they are susceptible to sensor drift, cross-sensitivity, and contact stability challenges, complicating their implementation in fluctuating field conditions [17].

In contrast, acoustic-based techniques provide a non-invasive and economical alternative by using sounds that reflect the mechanical and compositional properties of the fruit [11,10]. These approaches require only minimal and low-cost HW such as a smartphone or simple audio recorder—and remain inherently consistent with the conventional "knock test" traditionally employed by farmers. By analyzing frequency and resonance variations, acoustic signals enable effective discrimination between immature and mature durians without extensive calibration procedures. This study integrates agricultural knocking practices with modern acoustic signal processing to classify durian ripeness, leveraging the correlation between interior tissue characteristics and the resultant sound signatures.

Recent studies have demonstrated the effectiveness of DL-based classification in autonomously extracting features from Mel-Frequency Cepstral Coefficients (MFCCs) [18]. CNNs have shown particular strength because of the strong representation capabilities of these models [11]. CNNs are especially useful for audio processing because they form hierarchical representations from raw or minimally processed inputs and thus capture local spectral and temporal patterns evident in MFCCs. Convolutional filters extract crucial elements such as formant structures and harmonic patterns, which are vital for discerning small variations in audio streams. Kharamat et al. [11] illustrated that a CNN-based framework can effectively distinguish subtle differences in knocking noises corresponding to various stages of durian ripeness. Unlike traditional ML methods, which rely heavily on manually designed features and often struggle to capture complex, non-linear correlations in audio data, CNNs autonomously acquire strong and abstract representations from the data itself [19]. This comprehensive learning ability allows CNNs to adapt to minor fluctuations in audio inputs [20], resulting in enhanced classification performance. While CNN-based approaches have achieved high classification accuracy for durian ripeness using magnitude spectrograms, their omission of phase-based features limits the capture of fine-grained temporal cues—information shown in speech and bioacoustic studies to improve model robustness and discriminative power. Since comprehensive auditory representations require both magnitude and phase, neglecting the phase may result in the loss of essential audio details, as demonstrated in prior studies: Liu et al. [21] showed that phase conveys essential temporal cues in audio perception—demonstrated in speech by voicing distinctions in stop consonants—that cannot be captured by amplitude information alone, and Paliwal et al. [22] demonstrated through human perception experiments that the short-time phase spectrum can contribute to speech intelligibility as much as the magnitude spectrum, particularly when appropriate analysis window functions are used. Pathonsuwan et al. [23] confirmed the practical value of phase-based features by integrating magnitude and phase within a weighted feature-fusion CNN framework, achieving superior performance in chicken health detection compared with magnitude-only baselines.

In sound processing, numerous phase-based features have been investigated to capture critical spectral details. One commonly utilized feature is the Group Delay (GD) function, defined as the negative derivative of the phase spectrum [24]. While GD proficiently resolves phase wrapping, it may generate significant spikes when the denominator approaches zero. To address this issue, Hegde et al. [25] proposed a revised GD function, leading to the creation of Modified Group Delay Cepstral Coefficient (MGDCC). By mitigating the impact of spectral zeros and noise-related distortions, MGDCC features have consistently delivered robust performance in speech-processing application, as they preserve fine-grained spectral information that complements magnitude-based representations. Madikeri et al. [26] investigated MGDCC features inside the whole variability space, demonstrating their effective scalability and the supplementary information they offer to MFCC. Likewise, Deng et al. [27] integrated MFCC and GD for the recognition of emotions in whispered speech; nevertheless, their superficial model constrained the acquisition of profound emotional representations. To overcome this issue, Wang et al. [28] enhanced MFCC, MGDCC, and Power-Normalized Cepstral Coefficient (PNCC) within a Deep Neural Network (DNN) for speech activity detection, surpassing both MFCC/PNCC-only and previous combined magnitude-phase approaches. Specifically, their phase-aware DNN evaluated multiple feature fusion modes, including dual-magnitude with single-phase (e.g., PNCC+MGDCC+ Instantaneous Frequency derivative) and dual-phase with single-magnitude (e.g., PNCC+MGDCC+ baseband phase difference), as well as a score-level combination of these systems. The best outcome reduced the Equal Error Rate (EER) from 23.70% with MFCC-only to 19.44%, thereby demonstrating that combining magnitude and phase features achieves more noise-robust voice activity detection. Moreover, Guo et al. [29] shown that the incorporation of magnitude, MGDCC, and relative phase information inside a CNN framework markedly enhances speech emotion recognition compared to methods utilizing only magnitude or phase information.

Inspired by these findings, this study introduces *KnockNet*, a novel method that fully leverages the complementary characteristics of magnitude and phase information for classifying durian maturity based on knocking sounds. KnockNet specifically incorporates MFCC and MGDCC features, thereby preserving phase-related cues. Furthermore, to improve upon traditional CNN architectures, we introduce a Local Channel Attention (LCA) mechanism to highlight salient feature channels and a Convolutional Mixer (ConvMixer) module that employs depthwise and Pointwise Convolution (PWConv) to capture both local and global feature representations. By collaboratively utilizing these components, KnockNet provides a more thorough and resilient method for classifying durian ripeness. This dual-stream CNN-ConvMixer architecture incorporates parallel magnitude (MFCC) and phase (MGDCC) channels with LCA to produce richer and discriminative embeddings, achieving improved classification accuracy and creating a new standard for non-

[4] Post-harvest losses in tropical fruits have been estimated to average between 15%–25% of production, with specific surveys in Peninsular Malaysia indicating mean durian losses of 16.62%

[5] Post-harvest losses for durian in Peninsular Malaysia were reported at 16.62%, with 37.75% of farms losing 1–10%, 34.75% losing 11–20%, and 22.25% losing more than 20%.

[6] Post-harvest fruit rots in durian contributed to losses of 10%–25% of harvested fruit.

Table 1
Overview of existing non-invasive techniques for durian maturity/ripeness classification.

Reference	Techniques	Classification Method	Advantages	Disadvantages
[6]	Sensor-Based	ANN	Non-destructive; low-cost; MCU-deployable; Accuracy at 94.5%, F1 91.5	Small dataset (9 fruits); drift; environment sensitivity; confusions between ripe and overripe
[7]	Vision-Based	CNN (LeNet-5, AlexNet, DuNet-12)	High accuracy (95.8%–98.96%); DuNet-12 100% prediction (controlled)	Sensitive to background/lighting; controlled setup; high compute
[8]	Vision-Based	Deep transfer learning (MobileNet-v2, VGG16/19; Inception-v2/v3; ResNet-50/101/152; MobileNet-v3)	4-class Monthong dataset (1,000 images); high Accuracy—MobileNet-v2 95.5%, VGG16/19 94.5%; pretrained backbones; non-destructive	Image sensitivity to background/lighting; moderate dataset size; some backbones weak (Inception-v2 81.5%); field generalization not verified
[9]	Spectroscopy-Based	PLS-DA vs. ML (ANN, SVM, KNN, Ensemble)	Inline NIR; good overall (ANN: accuracy 85.3%, F1 85%, AUC 0.92)	Imbalanced data (immature ~40%); spectrometer + calibration
[10]	Acoustic/Vibration-Based	ANN	Low-cost; sound+vibration fusion improves accuracy (~70.6%); labeling support	Vibration-only < 10%; knocking realism
[11]	Acoustic-Based	CNN	Simple HW; Val 90.78%, Test 89.47%; 3-class	Limited data (30 fruits)
[12]	Spectroscopy-Based	LDA, SVM, KNN	Non-destructive; up to 100% (LDA+SWNIR); robust rind/stem; QC ready	Needs NIR spectrometer; preprocessing/calibration; higher cost
[13]	Spectroscopy-Based (Thermal)	PLS, PCA, LDA, kNN, SVM	Non-destructive; SVM 97%; correlates with SSC/firmness/pH/color	Thermal camera + calibration; ambient sensitivity; higher cost
[14]	Sensor-Based	PCA	Aroma-based discrimination (immature/ripe/over-ripe); non-destructive	Sensor drift; calibration complexity; cross-sensitivity
[15]	Sensor-/Spectroscopy (Review)	PLS, PCR, LDA, PCA, SVM, ANN (reviewed)	Broad non-destructive overview; Vis/NIR effective (SSC, acidity, firmness); fast, multi-attribute	Environment sensitivity; equipment cost; model transfer/robustness issues
[16]	Sensor-Based (EIS)	PLSR, Stepwise Reg., Discriminant	Minimally destructive; DM prediction (R = 0.83, RMSEP 4.25%); class accuracy 83.3–86.7%	Electrode insertion; moderate accuracy; polarization/contact resistance

Notes: Artificial Neural Network (ANN); Microcontroller Unit (MCU); F1-score (F1); Partial Least Squares Discriminant Analysis (PLS-DA); Machine Learning (ML); Support Vector Machine (SVM); k-Nearest Neighbors (KNN); Hardware (HW); Linear Discriminant Analysis (LDA); Short-Wave Near-Infrared (SWNIR); Partial Least Squares (PLS); Principal Component Analysis (PCA); Principal Component Regression (PCR); Visible and Near-Infrared (Vis/NIR); Partial Least Squares Regression (PLSR); Dry Matter (DM); Root Mean Square Error of Prediction (RMSEP).

invasive durian maturity assessment.
The key contributions of this study are outlined as follows:

1. **Architectural novelty:** We propose KnockNet, a dual-stream architecture that processes MFCC (magnitude) and MGDCC (phase) features before fusing them, thereby exploiting their complementary benefits. A streamlined ConvMixer backbone enables the integration of both local and global spectral–temporal representations into enhanced audio embeddings.
2. **Attention and efficiency:** Convolutional layers with LCA emphasize local spectral–channel cues (fine-resolution details), whereas a streamlined ConvMixer block captures global temporal–spectral dynamics. Their embeddings are fused to yield a richer representation of the knocking event than a conventional CNN.
3. **Evaluation:** Using a dataset of 52 Monthong fruits (2,918 knocks) collected in Chanthaburi Province, Thailand, and evaluated via five-fold cross-validation, KnockNet achieved $96.34 \pm 0.40\%$ accuracy (Precision/Recall/F1: 96.44/96.34/96.34%), surpassing MFCC-only, CNN, CNN-Long Short-Term Memory layers (CNN-LSTM), and SVM baselines.

2. Methodology

This section outlines each step of the processing pipeline, including dataset collection, feature extraction, and the proposed KnockNet architecture, which connects dual CNN streams with a ConvMixer module for global modeling.

2.1. Dataset collection and preprocessing

Immature, mature, and ripe samples of the 'Monthong' durian variety were collected from an orchard in the Khao Khitchakut area of Chanthaburi, eastern Thailand. A total of 52 samples were gathered, and their knocking sounds were recorded with a Sony ICD-PX470 digital voice recorder in MP3 format at 128 kbps. Fig. 1 illustrates the sampling procedure used to record knocking sounds from individual fruits. We converted the audio files into mono and downsampled them to 16 kHz for analysis to save file size, eliminate stereo channel redundancy, and ensure uniformity in spectrum analysis [30]. For each fruit, we tapped its center while the recorder captured the resulting sounds, producing a comprehensive database encompassing different phases of fruit maturation. The maturity phases were categorized as immature (82–106 Days After Anthesis, DAA), mature (113–148 DAA), and ripe, as suggested in [31].

A total of 1,924 sounds were recorded from fruits at early maturity stages with DAAs of 82, 84, 85, 86, 87, 88, 89, 99, and 106 (immature). DAAs 82–99 were measured on the same set of eight fruits; an additional four fruits were measured at DAA 106 to complete the immature cohort (see Table 2 note). An additional 637 sounds were recorded from mature fruits with DAAs of 113, 120, 127, 134, 141, and 148. Finally, 357 knocking sounds were recorded from ripe durians during post-harvest curing. These ripe samples were obtained from the same DAA levels listed for the mature cohort, but after the fruits underwent the described curing process. To simulate ripening and identify the "ready-to-eat" stage, 20 fruits (distinct from the 40 fruits sampled for the



(a) Durian tree in the experimental orchard.



(b) Handheld recording during manual knocking.

Fig. 1. Sampling procedure: (a) Experimental durian tree. (b) Handheld recording during manual knocking to acquire knock signals from individual fruits.

mature cohort) were selected for curing. Table 2 presents a summary of the maturity and ripeness data for the 'Monthong' durian variety.

In addition to DAA, we measured the Dry-weight percentage (DW%) of all mature durians used in this study to re-confirm physiological maturity. The DW% is computed as

$$DW\% = 100 \times \frac{W_{\text{dry}}}{W_{\text{fresh}}}, \tag{1}$$

where W_{dry} and W_{fresh} denote the pulp weights after drying and before drying, respectively. Equation (1) is explicitly referenced here and was used to determine the maturity thresholds reported below.

We defined "mature" as $DW\% \geq 32\%$ following prior work on the 'Monthong' cultivar [7,32,33]; this cultivar-specific threshold is commonly reported for Monthong and marks the onset of physiological maturity. In practice, values below $\sim 32\%$ typically indicate immature fruit that lacks sufficient DM for acceptable eating quality after curing, which explains our use of 32% in this study. Table 2 presents the per-DAA counts and totals; the table footnote has been updated to clarify the difference between unique fruits and total knocking sounds (repeated measures).

Table 2
Durian maturity and ripeness data for the *Monthong* variety.

Immature (82–106 DAA)				
DAA	Fruits	Sounds		
82	8†	189		
84	8†	173		
85	8†	189		
86	8†	191		
87	8†	342		
88	8†	398		
89	8†	283		
99	8†	123		
106	4	36		
Immature totals	**12‡**	**1,924**		
	Mature (113–148 DAA)		Ripe (113–148 DAA)	
DAA	Fruits	Sounds	Fruits	Sounds
113	4	75	4	72
120	4	69	3	64
127	8	96	1	25
134	8	128	4	64
141	8	145	4	66
148	8	124	4	66
Totals	**40**	**637**	**20**	**357**

Notes: DAA 82–99 were repeated measurements on the same 8 fruits (†). An additional 4 fruits were sampled at DAA 106, giving 12 unique immature fruits overall (‡) and 1,924 cumulative knocking sounds. The mature/ripe cohort (113–148 DAA) is separate from the immature cohort.

2.2. Feature extraction

Two complementary feature representations have been obtained from the knocking sounds: *magnitude* features utilizing MFCC and *phase* features employing MGDCC.

2.2.1. MFCC extraction

MFCC provides a compact representation of an audio signal's spectral envelope in a manner that corresponds with human auditory perception [34]. In our implementation, we apply a standard pre-emphasis filter, parameterized by a coefficient α that controls the amount of high-frequency emphasis, segment the signal into short, overlapping frames using a conventional frame/hop scheme, and apply a Hamming window prior to computing the Short-Time Fourier Transform (STFT) and the linear power spectrum $|X[k]|^2$. The extraction procedure begins with the application of a bank of Mel-scaled triangular filters to the signal's power spectrum, producing filter energies. The energies are logarithmically compressed to mimic the nonlinear response of the human ear, followed by a Discrete Cosine Transform (DCT) to decorrelate and compress the filter bank outputs [35].

Let $H_n[k]$ denote the n-th Mel filter and E_n its linear (power-domain) energy,

$$E_n = \sum_{k=0}^{K-1} |X[k]|^2 \, H_n[k], \qquad n = 1, \ldots, N_f,$$

which we convert to log energies $S_n = \log(E_n + \varepsilon)$ (natural log; $\varepsilon > 0$ avoids $\log 0$). Here, N_f denotes the number of Mel filters used to compute the log Mel energies (not the number of time frames).

Let S_n denote the logarithmically compressed energy of the n-th filter, where $n \in \{1, 2, \ldots, N_f\}$ and N_f is the total number of filters. The MFCC is computed as follows:

$$c_i = \sum_{n=1}^{N_f} S_n \cos\left(i\,(n-0.5)\,\frac{\pi}{N_f} \right), \quad i = 1, \ldots, M, \tag{2}$$

where c_i denotes the i-th MFCC coefficient and M is the number of coefficients retained. This DCT-based transformation efficiently captures the dominant characteristics of the spectral envelope while reducing redundancy in the representation.

2.2.2. MGDCC extraction

We utilize MGDCC to obtain phase information. GD is defined as the negative derivative of the phase spectrum with respect to frequency. The GD function is given by

$$\tau_x(\omega) = \frac{X_R(\omega)Y_R(\omega) + X_I(\omega)Y_I(\omega)}{|X(\omega)|^2}, \tag{3}$$

where $X(\omega)$ is the Fourier transform of the signal $x[n]$ with real and imaginary parts $X_R(\omega)$ and $X_I(\omega)$, and $Y(\omega)$ is the Fourier transform of $n\,x[n]$ with real and imaginary parts $Y_R(\omega)$ and $Y_I(\omega)$.

Wrapped phase introduces $\pm 2\pi$ jumps; differentiating with respect to frequency removes these, revealing the true phase slope. The GD cross-spectrum estimates this derivative without explicit unwrapping, with spikes near spectral zeros handled by the modified formulation.

To stabilize the estimate near zeros of $X(\omega)$, we replace the denominator by a cepstrally smoothed spectrum $S(\omega)$ and add a compression factor γ:

$$\tau(\omega) = \frac{X_R(\omega)Y_R(\omega) + X_I(\omega)Y_I(\omega)}{|S(\omega)|^{2\gamma}}, \qquad \gamma \in (0, 1]. \tag{4}$$

We then apply a sign-preserving nonlinearity to control dynamic range:

$$\tau_m(\omega) = \operatorname{sgn}\left(\tau(\omega)\right) |\tau(\omega)|^{\alpha}, \qquad \alpha \in (0, 1]. \tag{5}$$

Applying a DCT across the frequency bins of $\tau_m(\omega)$ yields the MGDCC feature vector. We adopt the same front-end configuration as in MFCC extraction, including pre-emphasis, short Hamming windows with overlap, and an Fast Fourier Transform (FFT) of suitable length. The procedure involves computing $X(\omega)$ and $Y(\omega)$, forming $\tau_x(\omega)$, stabilizing it with $|S(\omega)|^{2\gamma}$, transforming to $\tau_m(\omega)$, and finally applying the DCT to obtain the MGDCC representation. Phase information may be crucial here because knock sounds propagate through heterogeneous layers of durian peel and pulp; these layers introduce subtle GD and phase distortions that magnitude-only features (MFCC) may overlook. Incorporating MGDCC thus provides complementary cues for discriminating maturity stages, particularly when spectral envelopes are similar but internal structural differences alter phase responses.

2.3. KnockNet architecture

In this section, we present *KnockNet*, a two-stage architecture for ripeness classification. Fig. 2 illustrates that Stage 1 concurrently processes two complementary acoustic features—MFCC (magnitude) and MGDCC (phase)—through parallel CNN branches with identical architectures. The two streams do not interact in early layers; they are combined only at a late-fusion layer using channel-wise concatenation, followed by a 1×1 convolution into a common latent space. In this context, *local* context refers to spatial patterns identified within each CNN branch, whereas *global* context pertains to cross-channel dependencies represented by the ConvMixer via PWConv followed by channel mixing. Stage 2 applies the ConvMixer module to the fused representation to extract global temporal–spectral relationships prior to final classification. The subsequent subsections detail the processing pipeline. First, MFCC and MGDCC features are extracted in parallel CNN branches, followed by feature fusion using ConvMixer for integrated representation learning.

2.3.1. Dual-stream CNN processing of MFCC and MGDCC

KnockNet simultaneously processes MFCC (magnitude) and MGDCC (phase) inputs. Let $\mathbf{x} \in \mathbb{R}^{B\times 1\times H\times W}$ be a single-channel input feature map. Each stream advances in the following approach:

1) Convolutional processing. We begin with a 2D convolutional layer that employs a kernel of size $k \times k$ to learn localized patterns from the input $\mathbf{x}$. This layer is parameterized by weights $\mathbf{W}_1$ and bias $\mathbf{b}_1$ and computes the intermediate feature map as:

$$\mathbf{x}_1 = \mathbf{W}_1 * \mathbf{x} + \mathbf{b}_1. \tag{6}$$

Subsequently, Batch Normalization (BN) is applied to stabilize the learning process, followed by a Rectified Linear Unit (ReLU) activation to introduce non-linearity. A max-pooling operation with pooling size k_p is then used to reduce the spatial dimensions, producing the pooled feature map $\mathbf{x}_1^p$. Here, $*$ denotes 2D convolution; $\star$ denotes transposed convolution (see Eq. (8)); and $\odot$ denotes elementwise multiplication.

2) Second convolution. The second convolutional layer further refines these features. With parameters $\mathbf{W}_2$ and $\mathbf{b}_2$, it processes the pooled output from the previous layer:

$$\mathbf{x}_2 = \mathbf{W}_2 * \mathbf{x}_1^p + \mathbf{b}_2. \tag{7}$$

Again, BN and ReLU activation are applied to normalize and enhance the extracted features. Finally, another max-pooling operation of size k_p is performed to further condense the spatial dimensions, yielding the final feature representation $\mathbf{x}_2^p$.

3) LCA. To emphasize salient channels, the LCA module now operates on the pooled feature map $\mathbf{x}_2^p$. A transposed convolution is applied to $\mathbf{x}_2^p$ using parameters $\mathbf{W}_t$ and $\mathbf{b}_t$. A transposed convolution with stride $s > 1$ upsamples feature maps of $\mathbf{x}_2^p$, thereby increasing spatial resolution so that subsequent average pooling aggregates information over a broader, de-aliased neighborhood when forming channel descriptors [36]. The output is then passed through an average pooling layer, and a sigmoid activation produces the attention map:

$$\mathbf{A} = \sigma\Big(\operatorname{AvgPool}(\mathbf{W}_t \star \mathbf{x}_2^p + \mathbf{b}_t)\Big), \tag{8}$$

where $\star$ denotes the transposed convolution operation and σ represents the sigmoid function. The attention map $\mathbf{A}$ is applied elementwise to $\mathbf{x}_2^p$ to yield the attended features:

$$\mathbf{x}_2^{\text{att}} = \mathbf{x}_2^p \odot \mathbf{A}. \tag{9}$$

4) Channel refinement. Subsequently, a convolutional layer with weights $\mathbf{W}_3$ and bias $\mathbf{b}_3$ refines the attended features, changing the channel dimension from C_2 to C_3; this may represent an increase, a decrease, or no change depending on the layer configuration:

$$\mathbf{x}_3 = \mathbf{W}_3 * \mathbf{x}_2^{\text{att}} + \mathbf{b}_3. \tag{10}$$

A Leaky ReLU activation is applied for additional nonlinearity, yielding a final tensor $\mathbf{x}_3$ with dimensions (B, C_3, H', W').

2.3.2. Feature fusion and ConvMixer

After the MFCC and MGDCC streams have finished their respective processing, each stream outputs a final feature map, denoted by

$$\mathbf{x}_3^{(\text{MFCC})} \quad \text{and} \quad \mathbf{x}_3^{(\text{MGDCC})},$$

respectively. Both tensors are of dimension $\mathbb{R}^{B\times C\times H'\times W'}$, where B denotes the batch size, C the number of channels, and (H', W') the spatial dimensions.

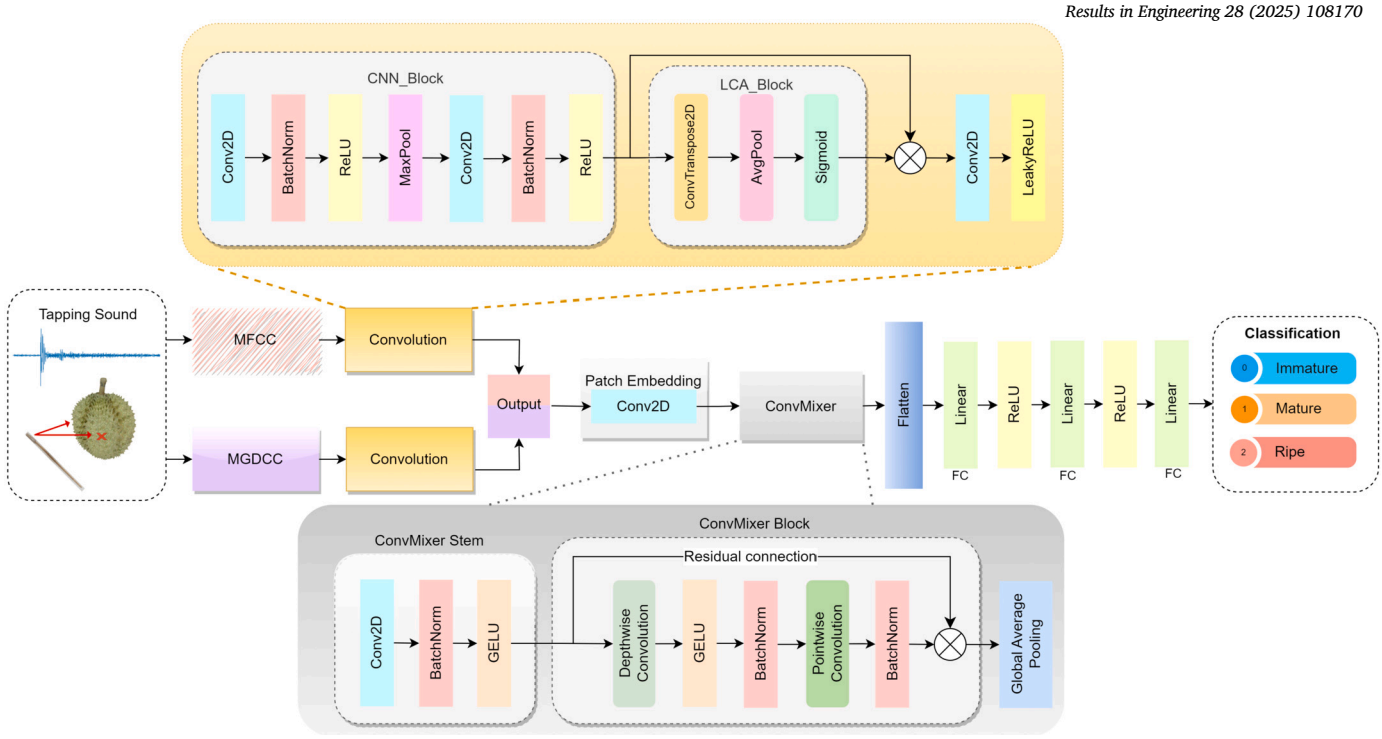


Fig. 2. Overview of the proposed KnockNet architecture.

1) Feature fusion. To combine the magnitude (MFCC) and phase (MGDCC) representations, these two feature maps are concatenated along the channel dimension:

$$\mathbf{x}_{\text{fusion}} = \text{Concat}\left(\mathbf{x}_3^{(\text{MFCC})}, \mathbf{x}_3^{(\text{MGDCC})}\right), \tag{11}$$

resulting in $\mathbf{x}_{\text{fusion}} \in \mathbb{R}^{B\times(2C)\times H'\times W'}$. This fused representation provides a richer feature set by integrating both magnitude and phase information.

2) Patch embedding. To reduce the spatial resolution and further increase the channel capacity, we employ a strided convolution (stride = 2). Let the weights and bias of this layer be $\mathbf{W}_e$ and $\mathbf{b}_e$, respectively. As expressed as follows:

$$\mathbf{z}_{\text{patch}} = \mathbf{W}_e * \mathbf{x}_{\text{fusion}} + \mathbf{b}_e, \tag{12}$$

where $*$ denotes 2D convolution. Consequently, $\mathbf{z}_{\text{patch}} \in \mathbb{R}^{B\times C_e\times \frac{H'}{2}\times \frac{W'}{2}}$, where C_e is the new number of channels.

3) ConvMixer stem. Before the main ConvMixer blocks, we apply a 1×1 "stem" convolution with weights $\mathbf{W}_{\text{stem}}$ and bias $\mathbf{b}_{\text{stem}}$, followed by BN and Gaussian Error Linear Unit (GELU):

$$\mathbf{z}_0 = \text{GELU}\left(\text{BN}\left(\mathbf{W}_{\text{stem}} * \mathbf{z}_{\text{patch}} + \mathbf{b}_{\text{stem}}\right)\right). \tag{13}$$

This produces $\mathbf{z}_0 \in \mathbb{R}^{B\times C_e\times \frac{H'}{2}\times \frac{W'}{2}}$, which is then fed into the ConvMixer blocks.

4)ConvMixer blocks. Let L be the total number of ConvMixer blocks. We denote by $\mathbf{z}_{l-1}$ the input to the l-th block and by $\mathbf{z}_l$ its output. Each block consists of three main steps: Depthwise Convolution (DWConv), PWConv, and a residual connection. Formally,

$$\mathbf{z}_l^{\text{dw}} = \text{BN}\left(\text{GELU}\left(\text{DWConv}(\mathbf{z}_{l-1})\right)\right), \tag{14}$$

$$\mathbf{z}_l^{\text{pw}} = \text{BN}\left(\text{PWConv}(\mathbf{z}_l^{\text{dw}})\right), \tag{15}$$

$$\mathbf{z}_l = \mathbf{z}_{l-1} + \mathbf{z}_l^{\text{pw}}, \tag{16}$$

where $\mathbf{z}_0$ from (13) is the input to the first block ($l = 1$). The DWConv captures spatial correlations within each channel, while the PWConv mixes features across channels. A residual connection facilitates stable gradient flow.

5) Global pooling and classification. After passing through L ConvMixer blocks, the final output $\mathbf{z}_L$ has dimensions $\mathbb{R}^{B\times C_e\times H''\times W''}$ (depending on how stride and kernel sizes are configured). To aggregate this spatial structure into a compact representation, we apply an adaptive average pooling:

$$\mathbf{v} = \text{Flatten}\left(\text{AdaptiveAvgPool}(\mathbf{z}_L)\right), \tag{17}$$

yielding $\mathbf{v} \in \mathbb{R}^{B\times D}$. A Fully Connected (FC) layer with optional dropout then projects $\mathbf{v}$ into the final prediction space:

$$\hat{\mathbf{y}} = \text{FC}(\mathbf{v}), \tag{18}$$

where $\hat{\mathbf{y}} \in \mathbb{R}^{B\times N_{\text{cls}}}$ encodes the logits or probabilities for the N_{cls} classes under consideration. The 1×1 convolution in the ConvMixer "stem" projects the concatenated patch embeddings into the internal channel dimension utilized across the ConvMixer blocks, enabling channel mixing and decreasing the parameter count relative to a larger kernel. It efficiently adjusts the integrated features to the ConvMixer's requisite channel width before performing depthwise operations.

3. Experimental setup

3.1. Training and validation protocols

In order to determine the most suitable window length for feature extraction, we evaluated frame sizes of 20, 40, 60, 100, and 200 ms with a 50% overlap during the computation of MFCCs while also investigating the impact of different numbers of coefficients (13, 20, 30, 40, 50, 60, 70, and 80). Empirical findings indicated that a 20 ms frame with 20 coefficients yielded the greatest classification accuracy for durian ripeness. Prior to framing, a pre-emphasis filter $y[t] = x[t] - \alpha x[t-1]$

Table 3
Per-fold split sizes (exact repetitions) and percentages relative to $N = 2{,}918$. Percentages rounded to one decimal place; counts are exact.

Setting	Train (% of N)	Validation (% of N)	Test (% of N)	Folds (count)
ML baselines (Train/Test = 80/20)				
Case A (Test = 584)	2,334 (80.0%)	–	584 (20.0%)	3
Case B (Test = 583)	2,335 (80.0%)	–	583 (20.0%)	2
DL models (Train/Val/Test = 60/20/20)				
Case A (Test = 584)	1,750 (59.9%)	584 (20.0%)	584 (20.0%)	3
Case B (Test = 583)	1,752 (60.0%)	583 (20.0%)	583 (20.0%)	2

with $\alpha = 0.97$ was applied to balance the spectral tilt and enhance high-frequency components relevant to knocking sounds. Therefore, 500 ms audio segments were analyzed via a STFT with 512 points, employing a 20 ms Hamming window with a 50% overlap. This configuration was used to extract both MFCC and MGDCC features. Through experiments conducted within the intervals $0 < \alpha \leq 1.0$ and $0 < \gamma \leq 1.0$, the parameters $\alpha = 0.4$ and $\gamma = 0.9$ were chosen for MGDCC, as they provided the highest possible classification accuracy for durian ripeness. Finally, all extracted features were subjected to mean–variance normalization [37] prior to inputting into the models, so ensuring uniform scaling across samples and enhancing the stability of the optimization process.

All experiments use stratified 5-fold cross-validation on the complete set of $N = 2{,}918$ repetitions (individual knock trials). All counts and splits encompass repetitions. ML baselines employ a Train/Test ratio of 80%/20% for each fold. DL models utilize a Train/Validation/Test split of 60%/20%/20% per fold, achieved by reserving a 20% test fold and dividing the remaining 80% into 75%/25% for training and validation, stratified within the training subset. Since N is not divisible by five, three folds yield a Test value of 584 and two folds yield a Test value of 583. Consequently, the sizes for ML training are 2,334 (when Test = 584) or 2,335 (when Test = 583); the DL per-fold Train/Val/Test distributions are 1,750/584/584 (three folds) and 1,752/583/583 (two folds). Table 3 presents precise counts and rounded percentages (to one decimal place).

The KnockNet model was constructed using PyTorch (v1.10.1) and trained on an NVIDIA RTX 3090 GPU (24 GB). The operating system was Ubuntu 14.04, operating on a workstation with 128 GB of RAM. The Adam optimizer was employed to minimize cross-entropy loss, utilizing the NVIDIA CUDA toolbox (v11.0) for GPU acceleration. An early stopping condition was implemented to avert overfitting: training was halted if the validation loss did not decline for 20 consecutive epochs. During training, performance was determined at each epoch, and model weights were saved only when validation loss decreased, leading to smooth convergence with a minimal train–validation discrepancy and ensuring the continuation of the best-generalizing model on the occurrence of early stopping. Fig. 3 shows the convergence curves of KnockNet across the five outer folds demonstrate consistent training dynamics, with early stopping triggered at the epochs where validation loss reached its minimum.

We performed an extensive grid search over learning rates $\{10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}, 10^{-6}\}$, convolutional kernel sizes $\{1\times1, 2\times2, 3\times3, 4\times4, 5\times5, 7\times7, 9\times9\}$, and batch sizes $\{8, 16, 32, 64\}$. The optimized KnockNet architecture is detailed layer-by-layer in Table 4, and the corresponding best-performing hyperparameters are listed in Table 5.

3.2. Evaluation metrics

To assess our model's performance, we use four common metrics: Precision, Recall, F1, and Accuracy. These metrics are defined as follows:

$$\text{Precision} = \frac{TP}{TP + FP},$$

$$\text{Recall} = \frac{TP}{TP + FN},$$

$$\text{F1} = 2\,\frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}},$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}.$$

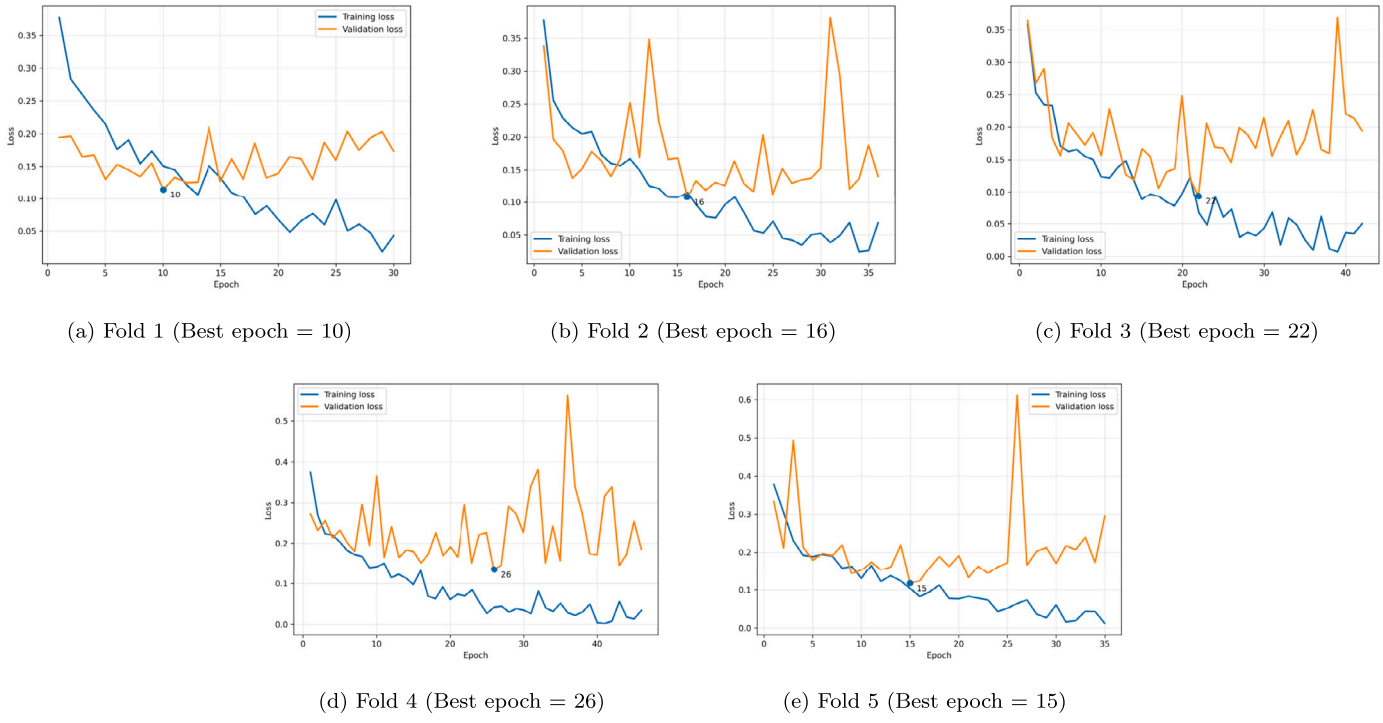
In these equations, TP denotes the number of true positives, FP the number of false positives, FN the number of false negatives, and TN the number of true negatives.

3.3. Compared methods

To thoroughly assess KnockNet, we compare its performance against various conventional ML and DL methodologies. All models utilize similar feature extraction, and training methodologies. The methods under comparison are outlined below:

- **KNN**: A non-parametric, instance-based classifier that assigns labels based on the majority vote of its k nearest neighbors.
- **SVM**: A classifier that finds an optimal separating hyperplane between classes, using kernel functions (e.g., radial basis function) to capture nonlinearity.
- **Decision Trees (DT)**: A tree-based method that recursively partitions the feature space via threshold-based splits, yielding an interpretable model.
- **Naïve Bayes (NB)**: A probabilistic classifier based on Bayes' theorem with an assumption of conditional independence among features, selected for its computational efficiency.
- **LDA**: A linear classifier that projects data onto a lower-dimensional subspace to maximize class separability under a Gaussian assumption.
- **CNN**: A DL model that learns spatially localized filters from time-frequency representations (e.g., MFCCs) to capture local patterns in the audio signals. Details on the CNN architecture are provided in the next section.
- **CNN–LSTM**: A hybrid architecture that uses the same CNN backbone as the CNN baseline to extract spectral features; the backbone outputs are pooled and fed to Long Short-Term Memory (LSTM) layers for temporal modeling, thereby combining spectral and temporal cues. Hyperparameters were tuned via exhaustive grid search on the training portion of each outer fold; stratified 3-fold splits of that training data were used to evaluate candidate settings, with macro-F1 as the selection metric. Candidate ranges were: hidden size $\{64, 128\}$, number of LSTM layers $\{1, 2\}$, dropout $\{0.2, 0.5\}$, and learning rate $\{10^{-3}, 10^{-4}\}$. The modal configuration across the five outer folds was: 1 LSTM layer, hidden size 128, dropout 0.5, and learning rate 10^{-3}.

All models were evaluated under *stratified* 5-fold cross-validation. Hyperparameters for the ML baselines were optimized via exhaustive grid search. Table 6 were evaluated using stratified 3-fold splits of each outer-fold training portion, with macro-F1 as the primary selection metric and accuracy used to break ties. The best hyperparameter setting from this inner search was then refit on the entire outer-fold training data and evaluated once on the held-out test set to avoid information leakage. Values quoted in the text (e.g., k=5 for KNN, C=1 and γ=`scale` for SVM) denote the *modal* (most frequently selected) choice across the five outer folds; per-fold selections are reported in Table 6. Preprocessing steps that require fitting (e.g., `StandardScaler`



(a) Fold 1 (Best epoch = 10) (b) Fold 2 (Best epoch = 16) (c) Fold 3 (Best epoch = 22)

(d) Fold 4 (Best epoch = 26) (e) Fold 5 (Best epoch = 15)

Fig. 3. Convergence curves of KnockNet across the five outer folds. Each subfigure highlights the epoch at which early stopping was applied (marked by a dot), corresponding to the minimum validation loss before overfitting began.

Table 4
Model configuration of the KnockNet.

Stage / Layer	Configuration	Output Shape
MFCC stream		
Input	–	$[B, 1, H, W]$
CNN_block	Conv2D($1 \to 16$, 3×3, $p=1$) + BN + ReLU MaxPool(2×2) Conv2D($16 \to 32$, 3×3, $p=1$) + BN + ReLU	$[B, 32, H/2, W/2]$
LCA_block	ConvTranspose2D($32 \to 32$, 3×3, $p=1$) + AvgPool + Sigmoid Elementwise multiply with input	$[B, 32, H/2, W/2]$
Conv2D	Conv2D($32 \to 64$, 3×3, $p=1$) + LeakyReLU	$[B, 64, H/2, W/2]$
MGDCC stream		
Input	–	$[B, 1, H, W]$
CNN_block	Conv2D($1 \to 16$, 3×3, $p=1$) + BN + ReLU MaxPool(2×2) Conv2D($16 \to 32$, 3×3, $p=1$) + BN + ReLU	$[B, 32, H/2, W/2]$
LCA_block	ConvTranspose2D($32 \to 32$, 3×3, $p=1$) + AvgPool + Sigmoid Elementwise multiply with input	$[B, 32, H/2, W/2]$
Conv2D	Conv2D($32 \to 64$, 3×3, $p=1$) + LeakyReLU	$[B, 64, H/2, W/2]$
Dual-Stream Fusion & ConvMixer		
Concat	Concatenate MFCC/MGDCC streams along channel dim	$[B, 128, H/2, W/2]$
Patch Embedding	Conv2D($128 \to 256$, 2×2, stride = 2)	$[B, 256, H/4, W/4]$
ConvMixer Stem	Conv2D($256 \to 256$, patch = 1) + GELU + BN	$[B, 256, H/4, W/4]$
ConvMixer Blocks $\times6$	DepthwiseConv2D($256 \to 256$, $k=9$, $p=4$) + BN + GELU PointwiseConv2D($256 \to 256$, 1×1) + BN Residual connection	$[B, 256, H/4, W/4]$
Global AvgPool	AdaptiveAvgPool2D(1×1)	$[B, 256, 1, 1]$
Flatten	–	$[B, 256]$
FC Head	Linear($256 \to 128$) + ReLU + Dropout(0.5) Linear($128 \to 64$) + ReLU Linear($64 \to 3$)	$[B, 3]$

Table 5
Primary hyperparameters for the KnockNet model.

Hyperparameter	Value
Batch size	16
Learning rate	0.001
Optimizer	Adam

for SVM/LDA) were computed only on the relevant training partition of each inner loop. All ML baselines were implemented using `scikit-learn` [38].

4. Results and discussion

4.1. Results with single-stream (SS)-CNN-ConvMixer models

To investigate the individual contributions of the LCA module and the ConvMixer block, we conduct ablation studies on the proposed SS-CNN-based ConvMixer (SS-CNN-ConvMixer) architecture using MFCC features. We define four variants of the SS-CNN-ConvMixer model by selectively including or omitting these components to quantify their impact on overall performance:

1. **SS-CNN-ConvMixer w/o LCA & ConvMixer:** Both LCA and ConvMixer are removed, leaving a standard CNN backbone. This model serves as the baseline CNN in our experiments.
2. **SS-CNN-ConvMixer w/o LCA:** The LCA module is excluded, but the ConvMixer remains.
3. **SS-CNN-ConvMixer w/o ConvMixer:** ConvMixer is excluded, keeping LCA to highlight salient channels.
4. **SS-CNN-ConvMixer:** This complete architecture includes both LCA and ConvMixer. Specifically, the *full* SS-CNN model consists of the CNN_block, LCA_block, Conv2D, Patch_Embedding, ConvMixer (ConvMixer Stem and ConvMixer Blocks), and a FC classification layer, with each component detailed in Table 4.

The effectiveness of the four SS-CNN-ConvMixer differences can be found in Table 7, evaluated based on Precision, Recall, F1, and Accuracy.

From Table 7, the following observation can be obtained:

- **Baseline CNN:** Removing both LCA and ConvMixer achieves around 94.64% Precision, 94.21% Recall, 94.29% F1, and 94.21% Accuracy. The absence of both modules limits the ability of the model to capture fine-grained temporal–spectral interactions.
- **SS-CNN w/o LCA:** Retaining the ConvMixer but omitting LCA yields improved Precision (95.15%) and Accuracy (95.00%), emphasizing the advantage of local–global feature mixing from ConvMixer.
- **SS-CNN w/o ConvMixer:** By contrast, having only the LCA module yields an F1 of 94.66% and an Accuracy of 94.62%. Although LCA effectively highlights the most important channels, broader contextual modeling of the ConvMixer appears essential for top performance.
- **SS-CNN Full:** Combining LCA with the ConvMixer (i.e., "SS-CNN Full") results in the best overall Accuracy (95.48%). The LCA module refines channel importance, while the ConvMixer captures more wide-ranging temporal and spectral dependencies—together producing a more robust representation.

4.2. Comparison of individual and stacking strategies

In this subsection, we evaluate the impact of magnitude-based MFCC and phase-based MGDCC features within a dual-stream framework on classification performance. Three configurations are compared: (1) **MFCC-only**, which applies the SS-CNN-ConvMixer architecture exclusively to MFCC features; (2) **MGDCC-only**, which employs the SS-CNN-ConvMixer architecture solely on MGDCC features; and (3) **MFCC+MGDCC (Stacking)**, corresponding to our proposed KnockNet that fuses both feature types. Table 8 summarizes the performance metrics of these three models.

From Table 8, the following observation of individual vs. stacking strategies can be obtained:

- **MFCC-SS-CNN-ConvMixer vs. MGDCC-SS-CNN-ConvMixer:** While both feature types capture critical acoustic cues, MFCC alone (95.48% Accuracy) outperforms MGDCC alone (93.28%). This indicates that magnitude-based features generally provide stronger discriminative power, although phase features still add value.
- **MFCC+MGDCC (Stacking):** Integrating magnitude and phase information achieves the highest Accuracy (96.34%). As shown in the confusion matrices in Fig. 4, this dual-stream approach reduces misclassification across ripeness classes, indicating that MFCC and MGDCC complement each other by capturing distinct aspects of the acoustic signal.

Fig. 4 shows confusion matrices for individual and stacking techniques. Visualizing these matrices allows us to identify class-level misclassifications and the overall efficacy of feature stacking. In Fig. 4(a), the MFCC-based model demonstrates exceptional accuracy for "Immature" samples (99.48%), although displays significant misunderstanding between "Mature" (88.70%) and "Ripe" (85.99%). In contrast, MGDCC alone (Fig. 4(b)) effectively recognizes "Immature" (99.32%) but has difficulties with "Ripe" (68.35%), frequently misclassifying these as "Mature." The integration of MFCC and MGDCC features (Fig. 4(c)) results in enhanced accuracy across all categories (e.g., 99.74% for "Immature," 90.27% for "Mature," 88.80% for "Ripe"), indicating that magnitude-based MFCC and phase-based MGDCC effectively capture complementary acoustic signals, thereby minimizing misclassifications overall. The test set shows an imbalance (Immature 65.9%, Mature 21.8%, Ripe 12.2%), indicating that the total accuracy is a weighted combination of each class's prevalence and recall [39]. For MFCC+MGDCC, recalls of 99.74%, 90.27%, and 88.80% with prevalences of 65.9%, 21.8%, and 12.2% yield contributions of 65.76, 19.71, and 10.86 percentage points, respectively, totaling 96.33%. This indicates that the much elevated Immature recall, due to its substantial occurrence, predominantly influences the total accuracy, elucidating why the stated 96.34% stays high despite diminished remembers for Mature and Ripe categories. To ascertain the significance of the observed gain, a paired t-test [40] conducted across the five outer folds comparing KnockNet (96.34%) with MFCC-only (95.48%) produced $t(4) = -2.80$, $p = 0.049$ (two-tailed), demonstrating statistical significance at $\alpha = 0.05$; the one-tailed test ($p = 0.024$) further substantiates that KnockNet significantly surpasses the SS baseline.

4.3. Comparison with baseline methods

To evaluate the effectiveness of our proposed *KnockNet*, we compare its performance against both classical ML and DL approaches. Table 9 highlights the classification results using various methods, including KNN, SVM, DT, LDA, NB, CNN-LSTM, CNN, and SS-CNN-ConvMixer models.

From Table 9, it is evident that ML classifiers trained on MFCC features achieve varying levels of accuracy, ranging from 87.29% (NB) to 93.83% (SVM). Among the ML models, SVM has superior performance, presumably owing to its capacity to establish appropriate decision boundaries in high-dimensional feature spaces. Nevertheless, these conventional ML techniques are fundamentally constrained by their manual feature extraction approach and their incapacity to model intricate temporal connections. Conversely, DL-based models exhibit enhanced performance, with CNN-LSTM and CNN surpassing all ML classifiers, with accuracy levels of up to 94.21%. The implementation of

Table 6
Hyper-parameter search space and selected values (modal across outer folds) for baseline ML systems.

Model	Hyper-parameter	Values tested	Final (modal)
SVM (Radial Basis Function, RBF/rbf)	C	$\{0.1, 1, 10, 100\}$	**1**
	γ	{"scale", "auto", 10^{-3}, 10^{-4}}	**scale**
	Kernel	{rbf}	**rbf**
DT	Criterion	{gini, entropy}	**gini**
	Max. depth	{None, 5, 10, 20}	**10**
	Min. split	{2, 5, 10}	**2**
	Min. leaf	{1, 2, 5}	**1**
NB (Gaussian)	var_smoothing	$\{10^{-9}, 10^{-8}, 10^{-7}\}$	**10^{-9}**
LDA	Solver	{svd}	**svd**
	Shrinkage	{None, "auto"}	**None**
	tol	$\{10^{-4}\}$	**10^{-4}**
KNN	k neighbors	$\{1, 3, 5, 7, 9, 11\}$	**5**
	Weights	{uniform, distance}	**uniform**
	Metric (p)	Minkowski $p \in \{1, 2\}$	**p=2**

Table 7
Performance of SS-CNN variants.

Model	Precision (%)	Recall (%)	F1 (%)	Accuracy (%)
Baseline CNN	94.64 ± 0.83	94.21 ± 0.81	94.29 ± 0.81	94.21 ± 0.81
SS-CNN-ConvMixer w/o LCA	95.15 ± 0.57	95.00 ± 0.79	94.96 ± 0.73	95.00 ± 0.79
SS-CNN-ConvMixer w/o ConvMixer	94.88 ± 0.71	94.62 ± 0.65	94.66 ± 0.66	94.62 ± 0.65
SS-CNN-ConvMixer	**95.69 ± 0.76**	**95.48 ± 0.61**	**95.52 ± 0.64**	**95.48 ± 0.61**

Table 8
Performance comparison of individual and stacking strategies.

Features	Precision (%)	Recall (%)	F1 (%)	Accuracy (%)
MFCC-only (Magnitude)	95.69 ± 0.76	95.48 ± 0.61	95.52 ± 0.64	95.48 ± 0.61
MGDCC-only (Phase)	93.16 ± 0.83	93.29 ± 0.79	93.17 ± 0.82	93.28 ± 0.79
MFCC+MGDCC (Stacking)	**96.44 ± 0.29**	**96.34 ± 0.40**	**96.34 ± 0.36**	**96.34 ± 0.40**

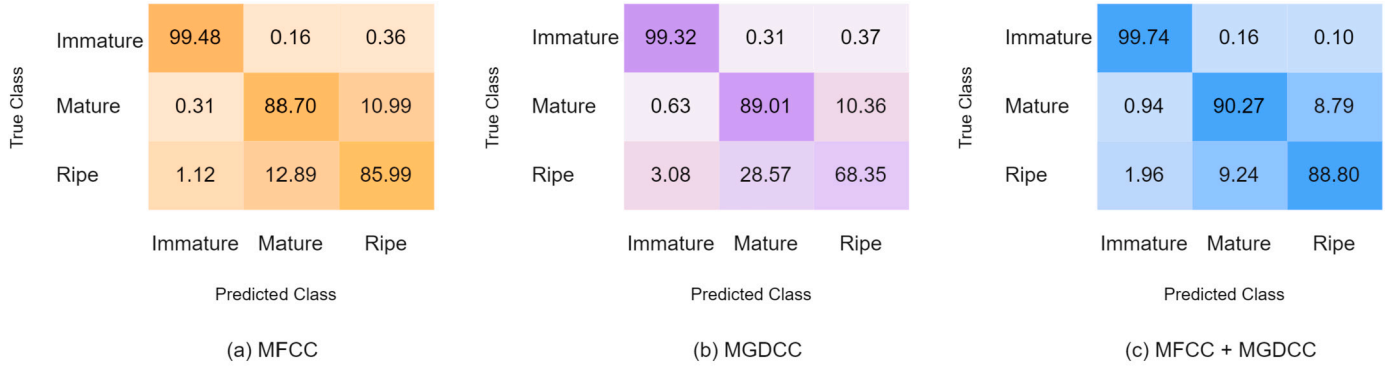


Fig. 4. Confusion matrices for various feature configurations: (a) MFCC, (b) MGDCC, and (c) MFCC+MGDCC.

Table 9
Comparison with other classification methods.

Method	Classifier	Precision (%)	Recall (%)	F1 (%)	Accuracy (%)
ML	KNN	93.55 ± 1.08	93.18 ± 1.03	93.16 ± 1.07	93.18 ± 1.03
	SVM	93.97 ± 0.80	93.83 ± 0.83	93.85 ± 0.85	93.83 ± 0.83
	DT	88.34 ± 1.43	88.18 ± 1.42	88.23 ± 1.41	88.18 ± 1.42
	LDA	89.52 ± 1.66	89.51 ± 1.35	89.45 ± 1.48	89.51 ± 1.35
	NB	89.22 ± 0.75	87.29 ± 0.95	87.45 ± 0.88	87.29 ± 0.95
DL	CNN-LSTM	94.78 ± 1.09	94.82 ± 1.13	94.75 ± 1.17	94.82 ± 1.13
	CNN	94.64 ± 0.83	94.21 ± 0.81	94.29 ± 0.81	94.21 ± 0.81
	SS-CNN-ConvMixer with MFCC	95.69 ± 0.76	95.48 ± 0.61	95.52 ± 0.64	95.48 ± 0.61
	SS-CNN-ConvMixer with MGDCC	93.16 ± 0.83	93.29 ± 0.79	93.17 ± 0.82	93.28 ± 0.79
	KnockNet (MFCC+MGDCC)	**96.44 ± 0.29**	**96.34 ± 0.40**	**96.34 ± 0.36**	**96.34 ± 0.40**

Table 10
Comparison with other classification methods: parameters, FLOPs, stored size (float32 / int8), and inference latency (CPU/GPU, batch size = 1). Sizes computed as size_float32 (MB) = $\frac{\text{params}\times 4}{1024^2}$ and size_int8 (MB) = $\frac{\text{params}\times 1}{1024^2}$. Latency is reported as mean ± standard deviation over 200 runs.

Classifier	Parameters	FLOPs	Size (float32)	Size (int8)	CPU latency	GPU latency
CNN-LSTM	347,331	13.97 M	1.32 MB	0.33 MB	8.19 ± 1.49 ms	0.68 ± 0.46 ms
CNN	116,931	6.74 M	0.44 MB	0.11 MB	5.23 ± 2.77 ms	0.51 ± 0.31 ms
SS-CNN-ConvMixer (MFCC)	733,702	39.62 M	2.79 MB	0.69 MB	12.02 ± 2.42 ms	1.56 ± 0.95 ms
SS-CNN-ConvMixer (MGDCC)	733,702	39.62 M	2.79 MB	0.69 MB	12.04 ± 2.08 ms	1.33 ± 0.91 ms
KnockNet (MFCC+MGDCC)	**831,107**	**51.48 M**	**3.17 MB**	**0.79 MB**	**16.13 ± 4.13 ms**	**2.36 ± 0.96 ms**

SS-CNN models enhances accuracy to 95.48%, largely attributable to the proficiency of the convolutional layers in extracting both local and global feature representations. Nevertheless, the complimentary nature of spectral magnitude and phase features is not completely utilized by SS designs, regardless of whether they employ MFCC or MGDCC. Our suggested *KnockNet*, which amalgamates MFCC and MGDCC within a dual-stream CNN architecture, markedly surpasses all baseline approaches, attaining the best accuracy of 96.34 ± 0.40%. The observed enhancement arises from the synergistic integration of magnitude-based (MFCC) and phase-based (MGDCC) features, which together offer a more comprehensive representation of the acoustic signals. While MFCC captures essential amplitude changes, MGDCC retains phase-related temporal dynamics, resulting in a more resilient feature representation. These results directly address previous shortcomings: MFCC-only pipelines ignore phase information, whereas handcrafted phase descriptors ignore magnitude changes. KnockNet's dual-stream design integrates magnitude and phase, while its ConvMixer backbone with LCA captures both local and global feature interdependencies. It allows various knock signatures by the promotion of channel-specific specialization. Collectively, these improvements constitute a principled solution that outperforms present techniques.

4.4. Analysis of model size and computational cost

Table 10 shows the parameter counts, Floating-point operations (FLOPs), model storage requirements, and inference latency (batch = 1) for the assessed DL classifiers. Classical ML baselines (KNN, SVM, DT, LDA, NB) are excluded as their computational and memory requirements are mostly influenced by data size and feature processing, rather than extensive learnable parameter sets.

CNN is the most lightweight model, including 116,931 parameters and 6.74 million FLOPs, requiring 0.44 MB (float32) / 0.11 MB (int8) of storage, and attaining 5.23±2.77 ms (CPU) and 0.51±0.31 ms (GPU) for batch-1 inference. CNN–LSTM comprises 347,331 parameters and 13.97 million FLOPs (1.32 MB / 0.33 MB), with a processing time of 8.19±1.49 ms on CPU and 0.68±0.46 ms on GPU. SS-ConvMixer variants (MFCC and MGDCC) utilize 733,702 parameters and 39.62 million FLOPs (2.79 MB / 0.69 MB), resulting in latencies of 12.02±2.42 ms / 1.56±0.95 ms and 12.04±2.08 ms / 1.33±0.91 ms (CPU / GPU), respectively. The proposed KnockNet (MFCC+MGDCC) integrates both streams, including 831,107 parameters and 51.48 million FLOPs (3.17 MB / 0.79 MB), with inference times of 16.13±4.13 ms (CPU) and 2.36±0.96 ms (GPU) per batch of one.

These results illustrate the trade-off between accuracy and cost: higher-performing multi-stream/global-context models require increased FLOPs, memory, and inference time. For deployment, simpler models (e.g., *CNN*) are preferable on severely constrained embedded CPUs; mid-complexity models (e.g., *CNN–LSTM*), SS *ConvMixer* are suitable for edge gateways or modest GPUs; and *KnockNet* is most appropriate for server or cloud inference or high-performance edge GPUs. Effective compression techniques, including pruning, post-training int8 quantization (about 4-fold storage reduction), and knowledge distillation, can significantly diminish resource demands while maintaining a majority of accuracy.

5. Conclusion

This paper proposed *KnockNet*, a dual-stream CNN-based ConvMixer that fuses magnitude (MFCC) and phase (MGDCC) representations using LCA and ConvMixer blocks to capture local and global temporal–spectral dependencies. Evaluation on a 52-sample Monthong dataset using stratified 5-fold cross-validation demonstrates that the proposed architecture delivers state-of-the-art performance for non-destructive durian ripeness classification. KnockNet achieved mean accuracy 96.34% (±0.40), F1 96.34% (±0.36), precision 96.44% (±0.29), and recall 96.34% (±0.40) across five outer folds. The full model requires approximately 0.831M parameters and 51.48M FLOPs. These quantitative indices provide concrete evidence that the dual-stream magnitude–phase fusion and the proposed attention/mixing components meaningfully improve classification performance compared with SS and classical baselines.

The results indicate a trade-off between accuracy and cost: multi-stream models improve classification but need more memory, computation, and inference time. To fulfill latency and energy budgets for deployment, we recommend HW-aware tuning and model compression (pruning, int8 quantization, knowledge distillation) or selective offloading to gateways/cloud. The main operational challenges are (i) operator variability and environmental noise, which can be mitigated with aggressive augmentation, robust pre-processing, and domain-adaptive fine-tuning; (ii) HW constraints, which can be addressed with HW-aware inference and model optimization; and (iii) cultivar/orchard variability, which can be addressed with larger, more varied datasets and domain-adaptation.

Future work will (a) increase the dataset across cultivars, regions and recording conditions, (b) enhance domain-adaptive and robust feature learning, (c) perform HW-aware compression and benchmarking on representative edge devices, and (d) investigate multimodal fusion (e.g., vibration sensors, vision) to improve operational reliability.

CRediT authorship contribution statement

Khomdet Phapatanaburi: Writing – review & editing. Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Khwanjit Orkweha:** Writing – review & editing, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Watcharakorn Pinthurat:** Methodology, Formal analysis, Conceptualization. **Chaiwat Makhonpas:** Methodology, Formal analysis, Conceptualization. **Wongsathon Pathonsuwan:** Methodology, Formal analysis, Conceptualization. **Patikorn Anchuen:** Methodology, Formal analysis, Conceptualization. **Monthippa Uthansakul:** Writing – review & editing, Formal analysis. **Peerapong Uthansakul:** Writing – review & editing, Supervision, Resources, Funding acquisition.

Declaration of generative AI and AI-assisted technologies in the writing process

We would like to declare that generative AI (ChatGPT-5) was used solely for assistance in checking and refining the English language in

this manuscript, including minor translations. The authors entirely generated the content, ideas, and findings presented in the manuscript without AI assistance. After language editing, the authors reviewed and validated the final version to ensure its accuracy and integrity.

Declaration of competing interest

The authors have declared no conflict of interest

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Data availability

Data will be made available on request.

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